



NTI MACHINE LEARNING PROJECT

Loan Approval Prediction Using Machine Learning

End-to-End Classification Pipeline with Hyperparameter Optimization

GitHub: <https://github.com/ahmedqataria99/Lendora>

Problem Statement & Project Objectives

Problem Statement

Predict whether a loan application is approved or rejected from applicant data — a key financial decision.

Manual credit checks cause major approval delays, high variance in reviewer subjectivity, and exposure to default risk. Intelligent modeling establishes consistent, data-backed thresholds.

Project Objectives

- BUILD A MACHINE-LEARNING CLASSIFICATION SYSTEM FOR LOAN APPROVAL PREDICTION
- PREPROCESS NUMERICAL AND CATEGORICAL APPLICANT DATA
- TRAIN MULTIPLE CLASSIFICATION ALGORITHMS
- TUNE THE MODELS USING GRIDSEARCHCV
- COMPARE THE MODELS USING EVALUATION METRICS
- SELECT THE BEST-PERFORMING MODEL
- PREPARE THE BEST MODEL FOR DEPLOYMENT

Dataset Overview

LOAN_DATA_NEW.CSV • 45,000 ROWS • 14 COLUMNS • TARGET: LOAN_STATUS (0 = REJECTED, 1 = APPROVED)

Numerical Features (8)

AGE
PERSON_INCOME
EMPLOYEE_EXPERIENCE
LOAN_AMOUNT
LOAN_INTEREST_RATE
LOAN_PERCENTAGE
CREDIT_HISTORY
CREDIT_SCORE

Categorical Features (5)

GENDER
EDUCATION
HOME_ONWERSHIP
LOAN_INTENT
PREVIOUS_LOAN

Exploratory Data Analysis

Analysis Performed

- DATASET INFO & DATA TYPES
- MISSING VALUES CHECK
- DUPLICATE ROWS CHECK
- DESCRIPTIVE STATISTICS
- CORRELATION ANALYSIS
- DISTRIBUTION ANALYSIS
- TARGET VARIABLE DISTRIBUTION

Key Findings

- NO MISSING VALUES
- NO DUPLICATE ROWS
- CLASS IMBALANCE IN THE TARGET
- PERSON_INCOME IS HIGHLY SKEWED
- EMPLOYEE_EXPERIENCE HAS MANY ZERO VALUES
- SOME EXTREME / UNREALISTIC VALUES EXIST

Target Distribution

01 REJECTED (0)

35,000 Records
77.78% of Total Dataset

Unapproved applications due to high debt burdens,
insufficient work history, or lower credit bands.

02 APPROVED (1)

10,000 Records
22.22% of Total Dataset

03 3.5x More Rejections than Approvals.

Data Preprocessing Pipeline

RAW DATA → NUMERICAL / CATEGORICAL SPLIT → IMPUTATION → SCALING / ONE-HOT ENCODING → PROCESSED DATA

NUMERICAL TRANSFORMATION:

Median Imputation + StandardScaler. Scaling matters for distance-based models like KNN and SVM.

CATEGORICAL TRANSFORMATION:

Most-Frequent Imputation + OneHotEncoder (handle_unknown='ignore', sparse_output=False) to convert them to numeric form.

PIPELINES & COLUMNTRANSFORMER:

Keep preprocessing organized, consistent, and leak-free.

Train / Test Split

80% TRAINING

36,000 TRAINING SAMPLES (80%)

RANDOM_STATE 42
STRATIFY Y TO PRESERVE THE ORIGINAL CLASS
DISTRIBUTION IN BOTH SETS.

20% TESTING

9,000 TEST SAMPLES (20%)

Held-out test set completely isolated from training to
ensure unbiased validation of accuracy metrics.

Machine Learning Models

Logistic Regression

PREDICTS APPROVAL PROBABILITY WITH A LINEAR DECISION BOUNDARY.

K-Nearest Neighbors

CLASSIFIES AN APPLICANT BY THE MAJORITY LABEL OF ITS NEAREST CASES.

Support Vector Machine

FINDS THE BOUNDARY THAT BEST SEPARATES THE TWO CLASSES.

Decision Tree

SPLITS DATA INTO RULES TO REACH AN APPROVE / REJECT DECISION.

Random Forest

COMBINES MANY DECISION TREES FOR STRONGER, STABLE PREDICTIONS.

AdaBoost

BOOSTS WEAK DECISION TREES BY FOCUSING ON THE HARD CASES.

Hyperparameter Tuning (Best Parameters - GridSearchCV)

Model	Best Parameters	Best CV
Logistic Regression	C=1, solver=lbfgs, class_weight=None	F1 0.761
KNN	n_neighbors=11, weights=distance, p=1, metric=minkowski	Acc 0.901
SVM	kernel=rbf, C=1, gamma=scale, class_weight=None	F1 tuned
Decision Tree	criterion=gini, max_depth=10, min_samples_leaf=5, min_samples_split=2	F1 0.803
Random Forest	n_estimators=200, max_depth=None, max_features=sqrt, min_samples_leaf=1, min_samples_split=10, class_weight=balanced	F1 0.827

Model Performance Comparison



Best Model: Random Forest (TEST ACCURACY 92.87%)

Classification Report

- Class 0 - Rejected:
Precision: 0.96 | Recall: 0.95 | F1: 0.95
- Class 1 - Approved:
Precision: 0.84 | Recall: 0.84 | F1: 0.84
- Overall Accuracy: 0.9287

Confusion Matrix & Best Params

CONFUSION MATRIX:

TN = 6672 | FP = 328
FN = 314 | TP = 1686

BEST PARAMS:

n_estimators = 200
max_depth = None
max_features = sqrt
min_samples_leaf = 1
min_samples_split = 10
class_weight = balanced

Boosting Comparison: AdaBoost vs Random Forest

AdaBoost Configuration

ADABOOST WITH A DECISION TREE BASE ESTIMATOR
(MAX_DEPTH=3), N_ESTIMATORS=100, LEARNING_RATE=0.1.

- ACCURACY: 91.39%
- F1-SCORE: 0.7910
- ROC-AUC: 0.9650

Benchmark Clarity

IN THIS BOOSTING SECTION THE NOTEBOOK REPORTS
RANDOM FOREST AT 93.00%, USED ONLY FOR THIS LOCAL
COMPARISON.

THE MAIN FINAL HELD-OUT TEST ACCURACY FOR RANDOM
FOREST REMAINS 92.87%. THE TWO FIGURES SHOULD NOT BE
CONFUSED.

Model Saving & Deployment

Saved Artifacts

- BEST_LOGISTIC_REGRESSION.JOBLIB
- BEST_KNN.JOBLIB
- BEST_SVM.JOBLIB
- BEST_DECISION_TREE.JOBLIB
- BEST_RANDOM_FOREST.JOBLIB
- PREPROCESSOR.JOBLIB

Deployment Architecture

SAVING THE TRAINED MODELS TOGETHER WITH THE PREPROCESSING OBJECT LETS THEM BE RELOADED AND REUSED LATER WITHOUT RETRAINING.

THE TRAINED MODEL AND PREPROCESSOR WERE SAVED FOR LATER DEPLOYMENT.

End-to-End Project Pipeline

DATASET

EDA

DATA
PREPROCESSING

TRAIN / TEST
SPLIT

BASELINE
MODELS

HYPERPARAMETER
TUNING

MODEL
EVALUATION

MODEL
COMPARISON

BEST MODEL
SELECTION

MODEL
SAVING

Thank You!

Questions?

LOAN APPROVAL PREDICTION

NTI MACHINE LEARNING PROJECT

Lendora: Smarter Lending. Better Decisions.

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